Major Project Report

A Study on Measuring Pedagogical Alignment in Online Learning Applications Using Evidence-Based Learning and Bloom's Taxonomy

Project Category: Research

Projexa Team Id - 26E3257

Submitted in partial fulfilment of the requirement of the degree of

BACHELOR OF COMPUTER APPLICATION

Semester-VI

to

K.R Mangalam University

by

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Under the supervision of

Supervisor Name : SHAQUIB HASAN

Department of Computer Science and Engineering
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K.R Mangalam University, Gurugram- 122001, India
May 2026

DECLARATION

We hereby declare that this project report is my original work and has not been submitted elsewhere. The information presented here is true and correct to the best of my knowledge.

Student Names:

Piyush Vats (2401201025)

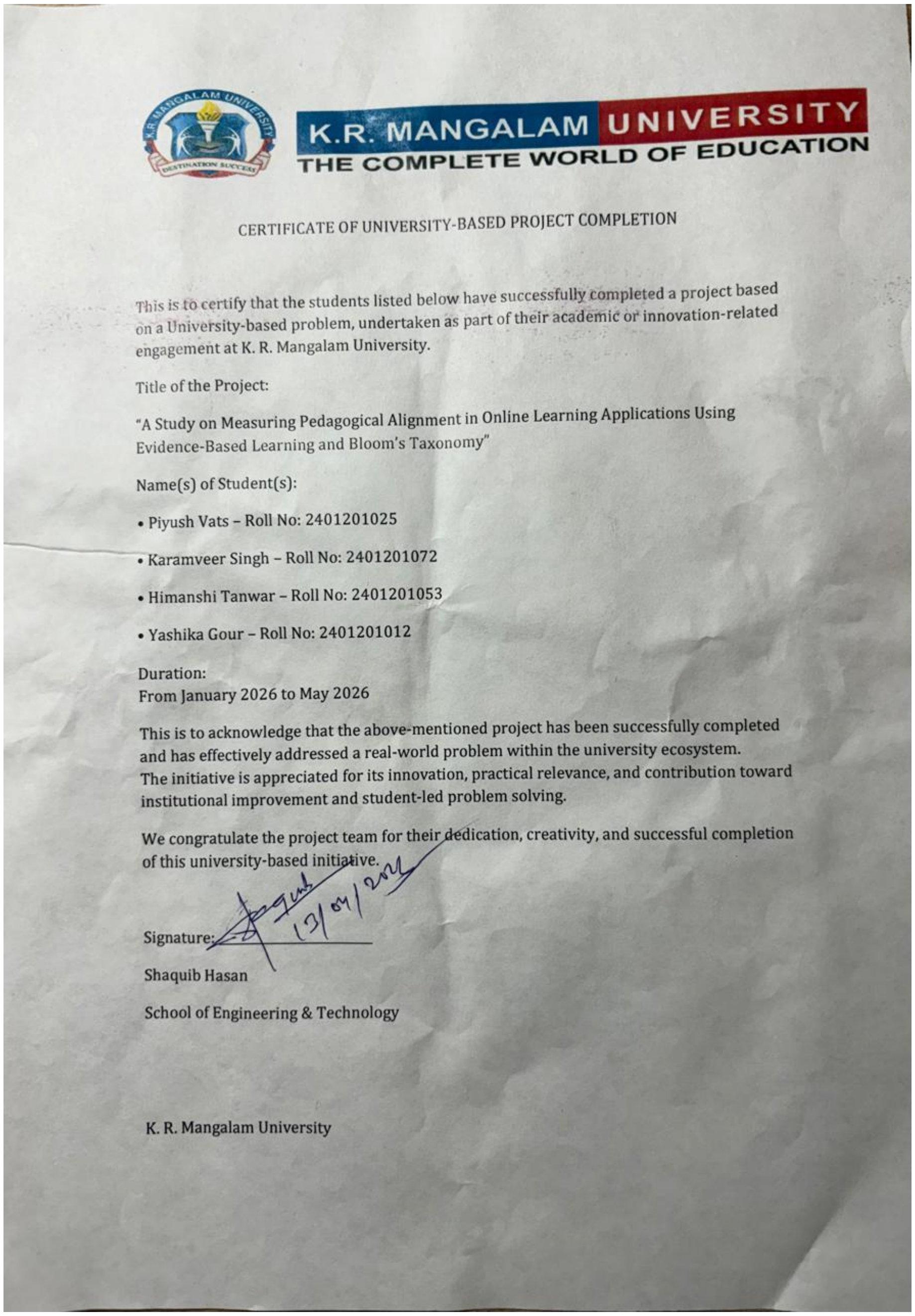
Karmveer Singh (2401201072)

Himanshi Tanwar (2401201053)

Yashika Gour (2401201012)

Date : 13 April 2026

PROJECT COMPLETION CERTIFICATE



K.R. MANGALAM UNIVERSITY
THE COMPLETE WORLD OF EDUCATION

CERTIFICATE OF UNIVERSITY-BASED PROJECT COMPLETION

This is to certify that the students listed below have successfully completed a project based on a University-based problem, undertaken as part of their academic or innovation-related engagement at K. R. Mangalam University.

Title of the Project:

"A Study on Measuring Pedagogical Alignment in Online Learning Applications Using Evidence-Based Learning and Bloom's Taxonomy"

Name(s) of Student(s):

- Piyush Vats – Roll No: 2401201025
- Karamveer Singh – Roll No: 2401201072
- Himanshi Tanwar – Roll No: 2401201053
- Yashika Gour – Roll No: 2401201012

Duration:
From January 2026 to May 2026

This is to acknowledge that the above-mentioned project has been successfully completed and has effectively addressed a real-world problem within the university ecosystem.
The initiative is appreciated for its innovation, practical relevance, and contribution toward institutional improvement and student-led problem solving.

We congratulate the project team for their dedication, creativity, and successful completion of this university-based initiative.

Signature: 13/04/2026

Shaquib Hasan

School of Engineering & Technology

K. R. Mangalam University

CERTIFICATE

This is to certify that the Project Synopsis entitled, A Study on Measuring Pedagogical Alignment in Online Learning Applications Using Evidence-Based Learning and Bloom's Taxonomy submitted by"**Piyush Vats(2401201025), Karamveer Singh(2401201072),Himanshi Tanwar(2401201053) and Yashika Gour(2401201012)"** to **K.R Mangalam University, Gurugram, India,** is a record of bonafide project work carried out by them under my supervision and guidance and is worthy of consideration for the partial fulfilment of the degree of **Bachelor of Technology** in **Computer Science and Engineering** of the University.

Internal supervisor : Shaquib Hasan

Date: 13^{th} April 2026

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ABSTRACT

In today's digital age, online learning applications have become a common tool for education, especially for young children. These applications are widely used because they are easy to access and provide interactive content that attracts learners. However, there is an increasing concern that many of these apps focus more on entertainment rather than actual learning.

Most learning apps use animations, rewards, and games to engage users, but they do not always help in developing strong understanding or thinking skills. This creates a gap between what users believe they are learning and what they actually learn. Currently, there is no proper system to evaluate whether these applications follow educational principles or not.

This research aims to analyze online learning applications using established frameworks such as Bloom's Taxonomy and Evidence-Based Learning (EBL). A structured evaluation method is developed to measure factors like engagement, cognitive level, and interaction. Different apps are studied and scored based on these criteria.

The result of this research is a framework that helps identify whether an app supports meaningful learning or just provides entertainment. This study can help parents, teachers, and developers make better decisions and improve the quality of digital education.

Chapter 1
Introduction

1. Background of the project

Technology has changed the way education is delivered. Online learning applications have become very popular, especially among children. These apps provide learning through games, videos, and interactive activities, making education more interesting.

Many parents now prefer learning apps because they are convenient and can be used anytime. These apps claim to improve skills like reading, writing, and problem-solving. However, there is a major concern about whether these apps actually provide real learning.

Most applications are designed to keep users engaged through rewards and visuals. While this increases usage, it does not always ensure understanding. Many apps focus only on basic tasks like memorization and repetition instead of developing deeper thinking skills.

To solve this issue, it is important to evaluate these apps using proper educational frameworks. Bloom's Taxonomy helps in understanding different levels of learning, while Evidence-Based Learning focuses on how effectively a learner is engaged.

This project aims to study learning apps in a structured way and check whether they support real learning or just create an illusion of learning.

2 MOTIVATION

The motivation behind this project comes from the increasing use of online learning apps in daily life. Children are spending more time on digital platforms for education, and parents trust these apps to support their learning.

However, during initial observation, it was found that many apps focus more on making learning “fun” rather than “effective.” Bright colors, sounds, and rewards make the app attractive, but they may not contribute to actual understanding.

This creates a serious problem because children may spend hours using these apps without gaining meaningful knowledge. Also, parents and teachers often rely on ratings and popularity, which do not reflect educational quality.

There is a need to understand what makes an app truly educational. This project is motivated by the idea of creating a system that can evaluate apps based on learning principles instead of just user experience.

The goal is to ensure that digital learning tools actually help students grow mentally and not just keep them busy.

Chapter 2
LITERATURE REVIEW

1. Review of existing literature

Various studies have been conducted in the field of digital learning and educational technology. Researchers have explored how learning apps can improve education and what factors make them effective.

One important concept is Bloom's Taxonomy, which explains different levels of learning such as remembering, understanding, and applying. Many studies show that effective learning requires higher-level thinking skills, not just memorization.

Another important concept is Evidence-Based Learning, which focuses on active participation, meaningful content, and interaction. Research suggests that students learn better when they are actively involved rather than passively consuming information.

Some studies also highlight that many modern apps focus more on engagement through rewards rather than learning outcomes. This creates a situation where apps look educational but do not actually support cognitive development.

Overall, the literature shows that there is a need for proper evaluation of learning apps using scientific methods.

2. GAP ANALYSIS

After reviewing existing research and applications, several gaps were identified.

Most existing studies focus on engagement and usability of apps but do not measure their educational effectiveness. There is a lack of proper frameworks to evaluate whether an app supports learning principles.

Another gap is that many apps do not focus on higher-order thinking skills. They mainly support basic learning like memorization, ignoring deeper understanding and problem-solving.

Also, there is no standard scoring system that can compare different apps based on their educational quality.

This project aims to fill these gaps by creating a structured evaluation framework that focuses on pedagogy and learning effectiveness.

3. PROBLEM STATEMENT

With the rapid growth of technology, online learning applications have become a major part of early education. These apps are widely used by children, parents, and teachers because they are easily accessible and provide interactive learning experiences. However, despite their popularity, there is a serious concern regarding their actual educational value.

Most online learning applications are designed to attract users through engaging elements such as animations, sound effects, rewards, and gamification techniques. While these features make the apps visually appealing and enjoyable, they do not necessarily contribute to meaningful learning. In many cases, children interact with these apps without developing a proper understanding of concepts, leading to passive learning rather than active learning.

Another important issue is that many learning apps do not follow established educational theories or frameworks. They mainly focus on basic tasks like memorization and repetition, ignoring higher-level cognitive skills such as understanding, application, analysis, and problem-solving. This limits the overall development of learners and reduces the effectiveness of these applications as educational tools.

Additionally, there is no standardized system available to evaluate the quality of these apps. Parents and teachers often rely on ratings, reviews, and popularity, which do not accurately reflect the educational effectiveness of the application. This makes it difficult to differentiate between truly educational apps and those that are primarily entertainment-based.

Therefore, there is a strong need to develop a structured and scientific approach to evaluate online learning applications. Such a system should focus on pedagogical alignment and help in identifying whether an app supports meaningful learning or not.

4. OBJECTIVES

The main objective of this research is to evaluate online learning applications and determine whether they provide real educational value or not. The study aims to analyze these applications using established learning theories and frameworks to understand their effectiveness.

The first objective is to study the current landscape of online learning applications, especially those designed for preschool and foundational level learners. This includes understanding their features, design patterns, and types of learning activities they provide.The second objective is to evaluate these applications using Bloom's Taxonomy. This helps in identifying the level of cognitive skills involved in the app, such as remembering, understanding, and applying concepts. The goal is to check whether the app supports higher-order thinking or not.The third objective is to analyze the applications based on Evidence-Based Learning principles. This includes evaluating whether the app promotes active engagement, meaningful learning, and interaction among users.

Another important objective is to identify the key features that contribute to effective learning in digital platforms. This includes analyzing elements such as feedback mechanisms, user interaction, and content structure.

The study also aims to develop a structured Pedagogical Evaluation Framework that can be used to measure the educational quality of learning applications in a systematic way.

Finally, the objective is to classify learning apps into different categories, such as high-quality and low-quality apps, based on their pedagogical alignment and effectiveness.

CHAPTER 3: METHODOLOGY

The methodology of this project follows a systematic and step-by-step approach to ensure accurate and reliable results. The entire process is divided into different phases.

The first phase is the Literature Review, where existing research and educational theories are studied. This helps in understanding the importance of pedagogical alignment and identifying key evaluation criteria.

The second phase is Framework Design. In this phase, a structured Pedagogical Evaluation Framework is developed based on Bloom's Taxonomy and Evidence-Based Learning principles. The framework includes different parameters such as engagement, interaction, and cognitive level.

The third phase is Data Collection. In this step, a list of online learning applications is selected from app stores. Relevant data such as features, activities, and user interaction is collected and recorded.

The fourth phase is Evaluation. Each application is analyzed using the developed framework. The apps are tested based on predefined criteria to understand their learning effectiveness.

The fifth phase is Scoring and Analysis. Each app is assigned a score based on its performance in different parameters. These scores are then analyzed to compare different applications.

The final phase is Result Generation. Based on the analysis, apps are categorized into high-quality and low-quality learning applications.

PLATFORM USED

In this project, various tools and platforms are used to collect, analyze, and evaluate data related to online learning applications. These tools are selected based on their simplicity, accessibility, and effectiveness in handling data.

Python is used as the primary programming language for data handling and analysis. It is chosen because it is easy to use and provides various libraries that support data processing and analysis. It also allows flexibility in handling different types of data.

Microsoft Excel or spreadsheet software is used for storing and organizing the collected data. It helps in maintaining records of different applications, their features, and evaluation scores in a structured format.

Online platforms such as Google Play Store and Apple App Store are used as data sources. These platforms provide access to various learning applications along with their descriptions, features, and user reviews.

Research papers and academic articles are also used to understand theoretical concepts such as Bloom's Taxonomy and Evidence-Based Learning. These sources help in building a strong foundation for the evaluation framework.

Overall, these tools help in simplifying the process of data collection, analysis, and evaluation, making the research more structured and effective.

Chapter 4
Implementation

EXPERIMENTAL SETUP

The experimental setup of this project involves selecting a set of online learning applications and evaluating them using a structured framework.

First, a dataset of different preschool and foundational level learning apps is created. These apps are selected from platforms like Google Play Store and Apple App Store based on their popularity and relevance.

Each app is installed and tested manually to observe its features, user interface, and learning activities. Special attention is given to how the app engages the user and whether it supports meaningful learning.

The evaluation is carried out using predefined parameters such as engagement level, interaction quality, and cognitive depth. Observations are recorded in a structured format using spreadsheets.

Multiple apps are compared based on their scores to identify patterns and differences. This helps in understanding which apps are more effective in supporting learning.

The experimental setup ensures that the evaluation process is consistent and reliable.

EVALUATION METRICS

The evaluation of learning applications is done using specific metrics that focus on both engagement and learning effectiveness.

The first metric is Active Engagement, which measures how actively the user participates in the learning process. Apps that require user interaction and decision-making score higher in this category.

The second metric is Meaningful Learning, which evaluates whether the app helps in understanding concepts rather than just memorizing information.

The third metric is Interaction Level, which checks whether the app provides feedback, guidance, or social interaction features.

Another important metric is Cognitive Level, based on Bloom's Taxonomy. This measures whether the app supports different levels of thinking such as remembering, understanding, and applying.

Each metric is given a score, and the total score determines the overall quality of the application.

Chapter 5

RESULTS AND DISCUSSIONS

The results of this research show that most online learning applications focus more on engagement rather than actual learning.

It was observed that many apps use animations, rewards, and sounds to attract users. While these features increase user interest, they do not contribute significantly to cognitive development.

Most applications mainly support basic learning levels such as remembering and understanding. Very few apps encourage higher-level thinking skills like problem-solving or analysis.

Another important finding is that many apps lack meaningful interaction. They do not provide proper feedback or guidance, which is important for effective learning.

However, some applications were found to follow educational principles and provide better learning experiences. These apps included interactive activities, logical progression, and feedback mechanisms.

Overall, the results highlight the need for improving the design of learning applications to focus more on education rather than entertainment.

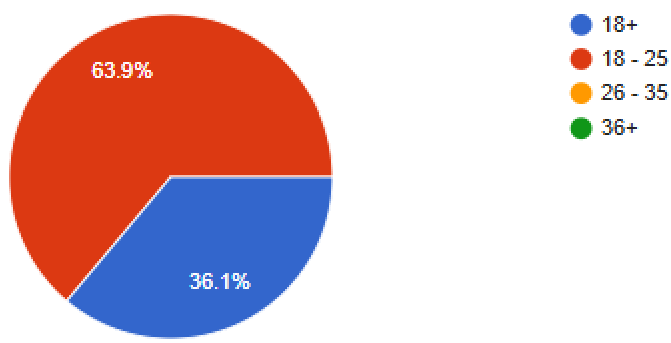
Google Form

https://docs.google.com/forms/d/e/1FAIpQLSfRF8pMYblp1PVKx2EyKBXxiUK5HrJ_S23jQgWGLZ18i-kNJQ/viewform

Outcomes of the form

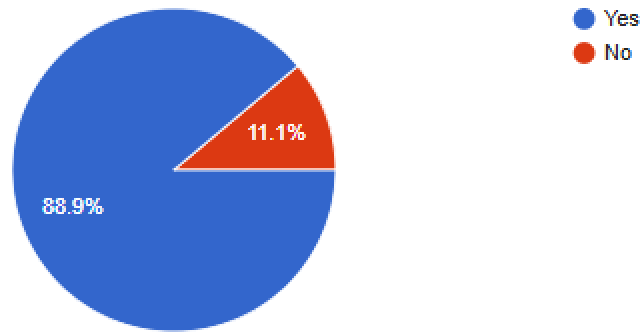
1. Age Group

36 responses



Have you used the learning apps for learning

36 responses

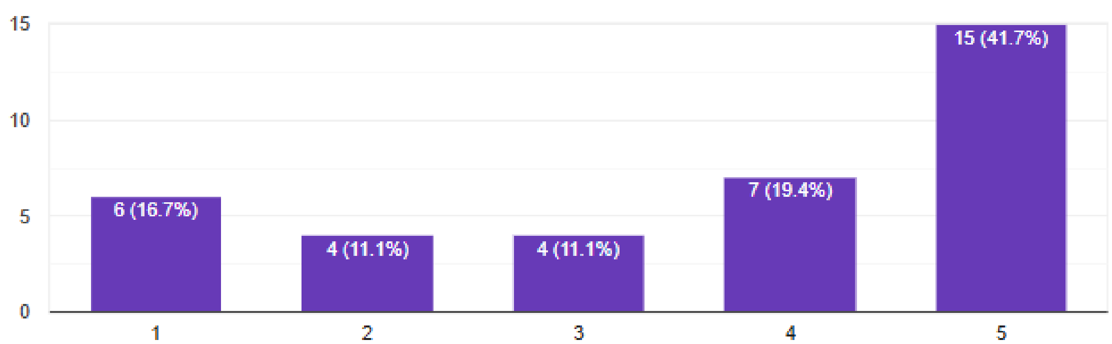


SECTION 3: LEARNING EFFECTIVENESS (GRAPH FRIENDLY 📊)

7. The app improves basic learning skills (reading, counting, etc. (1 = Strongly Disagree, 5 = Strongly Agree)

Copy chart

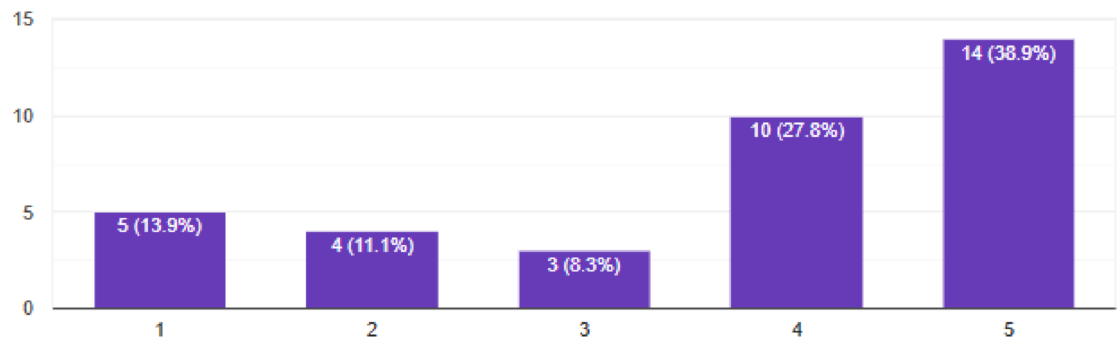
36 responses



8. The app helps in understanding concepts clearly .
(1 = Strongly Disagree, 5 = Strongly Agree)

Copy chart

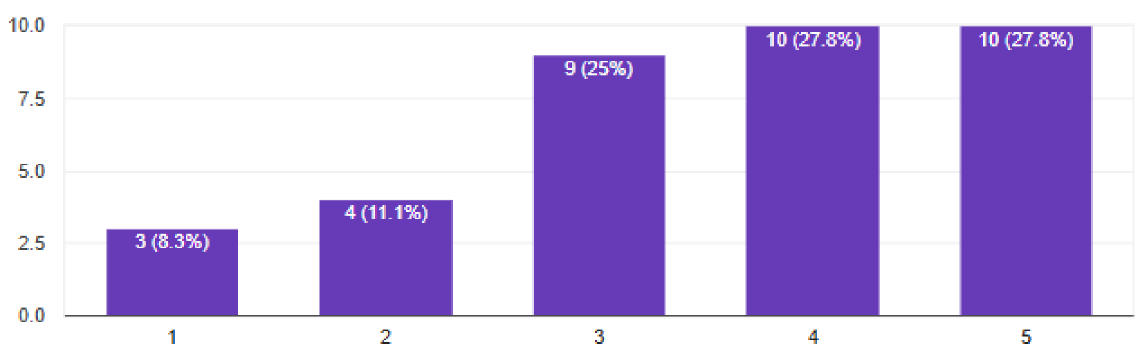
36 responses



9. The app encourages problem-solving skills
(1 = Strongly Disagree, 5 = Strongly Agree)

Copy chart

36 responses

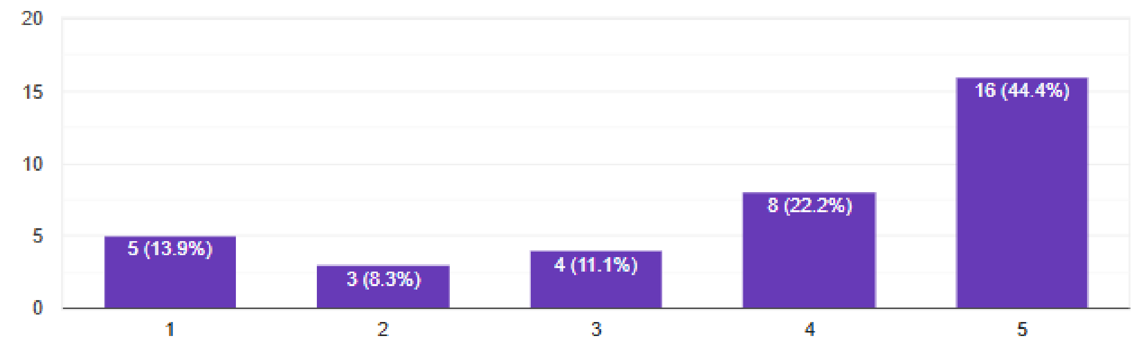


SECTION 4: PEDAGOGICAL ALIGNMENT

11. Learning objectives are clearly defined in the app
(1 = Strongly Disagree, 5 = Strongly Agree)

Copy chart

36 responses

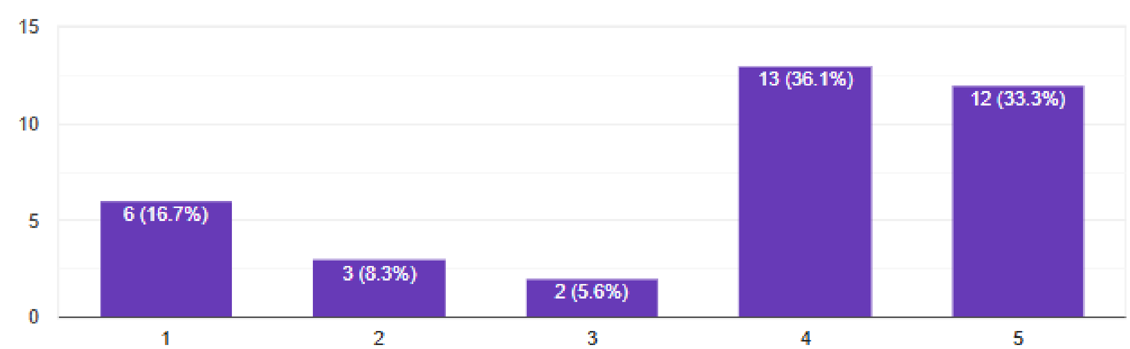


SECTION 7: BLOOM'S TAXONOMY

22. The app helps in remembering concepts

Copy chart

36 responses



Chapter 6

FUTURE WORK

This research concludes that although online learning applications are widely used, many of them do not provide effective learning experiences. They focus more on keeping users engaged rather than helping them understand concepts.

The study highlights the importance of evaluating apps using proper educational frameworks such as Bloom's Taxonomy and Evidence-Based Learning. The developed evaluation framework helps in identifying the strengths and weaknesses of different applications.

This research can help parents and teachers choose better learning tools for children. It also provides useful insights for developers to design apps that focus on meaningful learning.

For future work, the project can be extended by including more applications in the dataset and performing detailed statistical analysis. Machine learning techniques can also be used to predict the effectiveness of apps based on their features.

Additionally, real-time testing with students can be conducted to validate the results and improve the evaluation framework.

REFERENCES

1. Research papers on Bloom's Taxonomy
2. Studies on Evidence-Based Learning
3. Google Scholar articles
4. Data collected from app stores

ANNEXURE I:

Plagiarism Declaration Certificate

Title of Work: A Study on Measuring Pedagogical Alignment in Online Learning Applications Using Evidence-Based Learning and Bloom’s Taxonomy
Submitted By: Piyush Vats
Institution: K. R. Mangalam University, Gurugram
Department: Bachelor of Computer Application,School of Technology and Engineering
Date of Submission: 13th April 2026

I hereby declare that the work entitled **”[A Study on Measuring Pedagogical Alignment in Online Learning Applications Using Evidence-Based Learning and Bloom’s Taxonomy]”** submitted for academic evaluation and research purposes is my original work. I confirm that:

- I have acknowledged and properly cited all sources, references, and data included in this work.
- This work does not contain any material previously published, written, or prepared by another person, except where due acknowledgment has been made.
- I understand that plagiarism is an academic offense and a violation of research ethics. Any breach of this declaration may lead to disciplinary action as per university policies.
- I have used appropriate referencing techniques and maintained academic integrity throughout this work.

I affirm that the submitted work has normal plagiarism less than 10% and free from AI content.

___

Name of Student: Piyush Vats
Date: 13th April 2026

ANNEXURE III:

Excel Sheet Data

URL:[https://docs.google.com/spreadsheets/d/15oUzgCPxQ8cItmrbbOoEoMM526Th214j77J-t1fdUoE/edit?resourcekey=&gid=1836980715 - gid=1836980715](https://docs.google.com/spreadsheets/d/15oUzgCPxQ8cItmrbbOoEoMM526Th214j77J-t1fdUoE/edit?resourcekey=&gid=1836980715#gid=1836980715)

Outcomes of Excel sheet

Form_Responses

1	Timestamp	1. Age Group	2 Role	Have you used the learning apps for learning	4. Which learning apps have you used?	5. How frequently are these apps used?	6. Average time spent
2	06/04/2026 20:12:57	18+	Student	Yes	Byju's	Weekly	More then 1 hour
3	06/04/2026 20:14:32	18+	Student	Yes	Byju's	Weekly	15 - 30 min
4	06/04/2026 20:14:48	18+		Yes	Byju's	Weekly	30 - 60 min
5	06/04/2026 20:19:38	18 - 25	Student	Yes	Byju's	Occasionally	More then 1 hour
6	06/04/2026 20:22:59	18 - 25	Student	Yes	Khan Academy Kids	Occasionally	15 - 30 min
7	06/04/2026 20:23:35	18 - 25	Student	Yes	Byju's	Weekly	15 - 30 min
8	06/04/2026 20:27:15	18+	Student	Yes	Photomath	Weekly	15 - 30 min
9	06/04/2026 20:29:13	18+	Student	Yes	Photomath, Byju's	Occasionally	15 - 30 min
10	06/04/2026 20:31:02	18 - 25	Student	Yes	Byju's	Daily	More then 1 hour
11	06/04/2026 20:32:17	18+	Student	Yes	Byju's	Weekly	15 - 30 min
12	06/04/2026 20:34:19	18+	Student	Yes	Khan Academy Kids	Weekly	15 - 30 min
13	06/04/2026 20:35:53	18 - 25	Student	Yes	Photomath	Weekly	15 - 30 min

1	7. The app improves basic learning skills (re	8. The app helps in understanding concepts (1 = Strongly Disagree, 5 = Strongly Agree)	9. The app encourages problem-solving skil (1 = Strongly Disagree, 5 = Strongly Agree	10. The app supports higher-order thinking (1 = Strongly Disagree, 5 = Strongly Agree)	11. Learning objectives are clearly def (1 = Strongly Disagree, 5 = Strongl
2	3	4	2	3	
3	5	5	5	5	
4	4	4	3	4	
5	5	5	5	5	
6	3	3	3	3	
7	4	4	4	4	
8	1	2	3	3	
9	2	5	5	5	
10	5	5	5	5	
11	1	1	3	3	
12	5	3	2	2	
13	5	5	5	5	

A1 Timestamp

	M	N	O	P	Q
1	12. Activities match the learning goals (1 = Strongly Disagree, 5 = Strongly Agr	13. Assessment/quiz features are effective (1 = Strongly Disagree, 5 = Strongly Agre	14. Content is structured and progressive (1 = Strongly Disagree, 5 = Strongly Agre	15. The app is engaging for children (1 = Strongly Disagree, 5 = Strongly Agre	16. The app is more entertaining than edu (1 = Strongly Disagree, 5 = Strongly A
2	1	1	1	1	
3	4	4	3	4	
4	4	5	4	4	
5	5	5	5	5	
6	3	3	3	3	
7	3	4	4	3	
8	5	5	3	3	
9	3	2	5	5	
10	4	5	4	5	
11	1	3	3	2	
12	4	4	4	5	
13	5	5	5	5	

A24 06/04/2026 20:56:24

Form_Responses

	A	B	C	D	E	F	G
1	Timestamp	1. Age Group	2 Role	Have you used the learning apps for learning	4. Which learning apps have you used?	5. How frequently are these apps used?	6. Average time spent
15	06/04/2026 20:41:06	18+	Student	Yes	Creta class	Weekly	15 - 30 min
16	06/04/2026 20:43:53	18+	Parent	Yes	Khan Academy Kids, Byju's	Daily	15 - 30 min
17	06/04/2026 20:47:02	18 - 25	Student	Yes	Khan Academy Kids	Occasionally	Less than 15 min
18	06/04/2026 20:51:46	18 - 25	Student	Yes	Creta class	Daily	30 - 60 min
19	06/04/2026 20:51:48	18 - 25	Student	Yes	Creta class	Daily	More then 1 hour
20	06/04/2026 20:53:33	18 - 25	Student	Yes	Khan Academy Kids	Daily	30 - 60 min
21	06/04/2026 20:53:38	18 - 25	Student	Yes	Photomath	Occasionally	30 - 60 min
22	06/04/2026 20:54:04	18+	Student	Yes	Photomath, Byju's	Daily	More then 1 hour
23	06/04/2026 20:55:03	18 - 25	Student	Yes	Photomath	Daily	30 - 60 min
24	06/04/2026 20:56:24	18 - 25	Student	Yes	Byju's	Daily	30 - 60 min

	I	J	K	L	M
1	e app helps in understanding concepts trongly Disagree, 5 = Strongly Agree)	9. The app encourages problem-solving skil (1 = Strongly Disagree, 5 = Strongly Agree	10. The app supports higher-order thinking (1 = Strongly Disagree, 5 = Strongly Agree)	11. Learning objectives are clearly define (1 = Strongly Disagree, 5 = Strongly A	12. Activities match the learning goals (1 = Strongly Disagree, 5 = Strongly Agr
15	5	3	2	2	4
16	5	3	4	4	5
17	1	1	4	2	1
18	4	5	5	4	4
19	5	4	5	5	5
20	4	4	4	5	4
21	2	2	2	2	1
22	5	3	4	4	3
23	4	4	4	5	5
24	5	5	4	5	4

Form responses 1

AH24:AI24 Nothing

	AD	AE	AF	AG	AH
1	erall rating of learning apps = Strongly Disagree, 5 = Strongly Agr	31. Do you think these apps replace traditio	32. What do you like most about learning ap	33. What improvements are needed?	34. Any suggestions for better educational
2	3	No			
3	10	Maybe	Gaining knowledge	The app should be more understandable	No
4	9	Maybe	That we get the most updated knowledge about tl	Should be more flexible	No
5	10	Yes	Bjyus	Every questions are well answered	Nope
6	3	No	Nothing	No	No
7	6	Maybe	They are quiet engaging	Learning material can be improved	Give examples to engage learning
8	5	Yes	They make learning interactive and fun with quizz	Reduce ads as they distract from learning.	Include gamification like rewards and badges to r
9	10	Yes	Learning with animation	Should be more engaging	Should provide free and reliable solutions
10	10	No	I like the flexibility and convenience of learning ap	Learning apps can improve by reducing dist	Educational apps should include more interactive
11	4	Yes	It's very effective	Nothing	No
12	6	No	I can learn anytime and anywhere at my own pace	Improve content quality and accuracy.	Provide personalized learning paths based on use
13	7	No	They make learning interactive and fun	Low ads	Add AI based tutor
				1. Reduce ads as they distract from lear 2. Improve content quality and accu 3. Add more interactive activities in 4. Make apps less expensive or pro	1. Include gamification like rewards and badg 2. Provide personalized learning paths ba 3. Add AI-based tutors for instant doubt s 4. Include real-world problem-solving acti

UI : Code

```
/* 1. Global Variables & Reset */
:root {
    --primary: #6c5ce7;
    --secondary: #a29bfe;
    --success: #55efc4;
    --warning: #f1c40f;
```

```css
    --error: #ff7675;
    --dark-bg: #1e272e;
    --glass: rgba(255, 255, 255, 0.1);
    --glass-heavy: rgba(255, 255, 255, 0.2);
}

body {
    font-family: 'Segoe UI', Arial, sans-serif;
    background: linear-gradient(135deg, #0f0c29, #302b63, #24243e);
    color: white;
    margin: 0;
    padding: 0;
    text-align: center;
    min-height: 100vh;
    overflow-x: hidden;
}

/* 2. Navigation & Header */
header {
    background: rgba(0, 0, 0, 0.4);
    backdrop-filter: blur(10px);
    padding: 15px 20px;
    border-bottom: 1px solid rgba(255, 255, 255, 0.1);
    position: sticky;
    top: 0;
    z-index: 1000;
}

.nav-container {
    display: flex;
    justify-content: space-between;
    align-items: center;
    max-width: 1200px;
    margin: auto;
}

.logo {
    font-size: 24px;
    font-weight: bold;
    color: var(--success);
}

nav a {
    text-decoration: none;
    color: #dfe6e9;
```

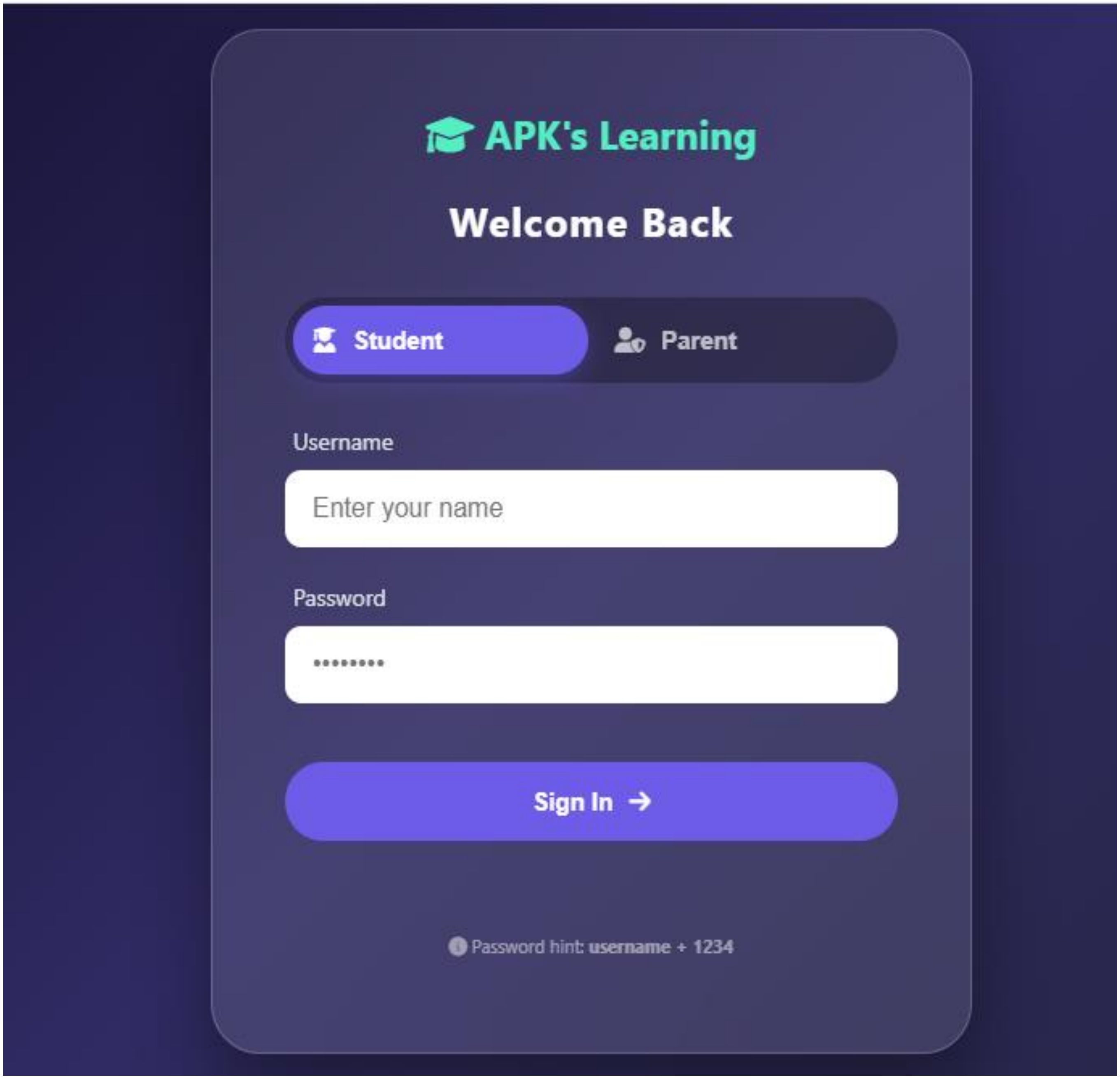
```css
    padding: 10px 15px;
    border-radius: 20px;
    transition: 0.3s;
    margin: 0 5px;
}

nav a.active {
    background: var(--primary);
    color: white;
}

/* 3. Gamification: XP Toast Notification */
.xp-toast {
    position: fixed;
    bottom: 25px;
    right: 25px;
    background: var(--warning);
    color: #2d3436;
    padding: 12px 25px;
    border-radius: 50px;
    font-weight: bold;
    box-shadow: 0 10px 20px rgba(0,0,0,0.3);
    z-index: 9999;
    animation: slideInRight 0.5s cubic-bezier(0.175, 0.885, 0.32, 1.275);
}

@keyframes slideInRight {
    from { transform: translateX(120%); }
    to { transform: translateX(0); }
}

/* 4. Interactive Flashcards */
.flashcard-container {
    perspective: 1000px;
    width: 100%;
    max-width: 400px;
    height: 180px;
    margin: 20px auto;
}

.flashcard {
    width: 100%;
    height: 100%;
    transition: transform 0.6s;
    transform-style: preserve-3d;
```

```css
    cursor: pointer;
    position: relative;
}

.flashcard.flipped {
    transform: rotateY(180deg);
}

.card-front, .card-back {
    position: absolute;
    width: 100%;
    height: 100%;
    backface-visibility: hidden;
    display: flex;
    align-items: center;
    justify-content: center;
    border-radius: 20px;
    padding: 25px;
    box-sizing: border-box;
    font-weight: 600;
    font-size: 1.1rem;
    box-shadow: 0 8px 16px rgba(0,0,0,0.2);
}

.card-front {
    background: linear-gradient(45deg, var(--primary), var(--secondary));
    color: white;
    border: 1px solid rgba(255,255,255,0.2);
}

.card-back {
    background: var(--success);
    color: #1e272e;
    transform: rotateY(180deg);
}

/* 5. FIX: Form Inputs */
input {
    width: 100%;
    padding: 12px 15px;
    border-radius: 10px;
    border: 2px solid transparent;
    font-size: 16px;
    box-sizing: border-box;
    transition: 0.3s;
```

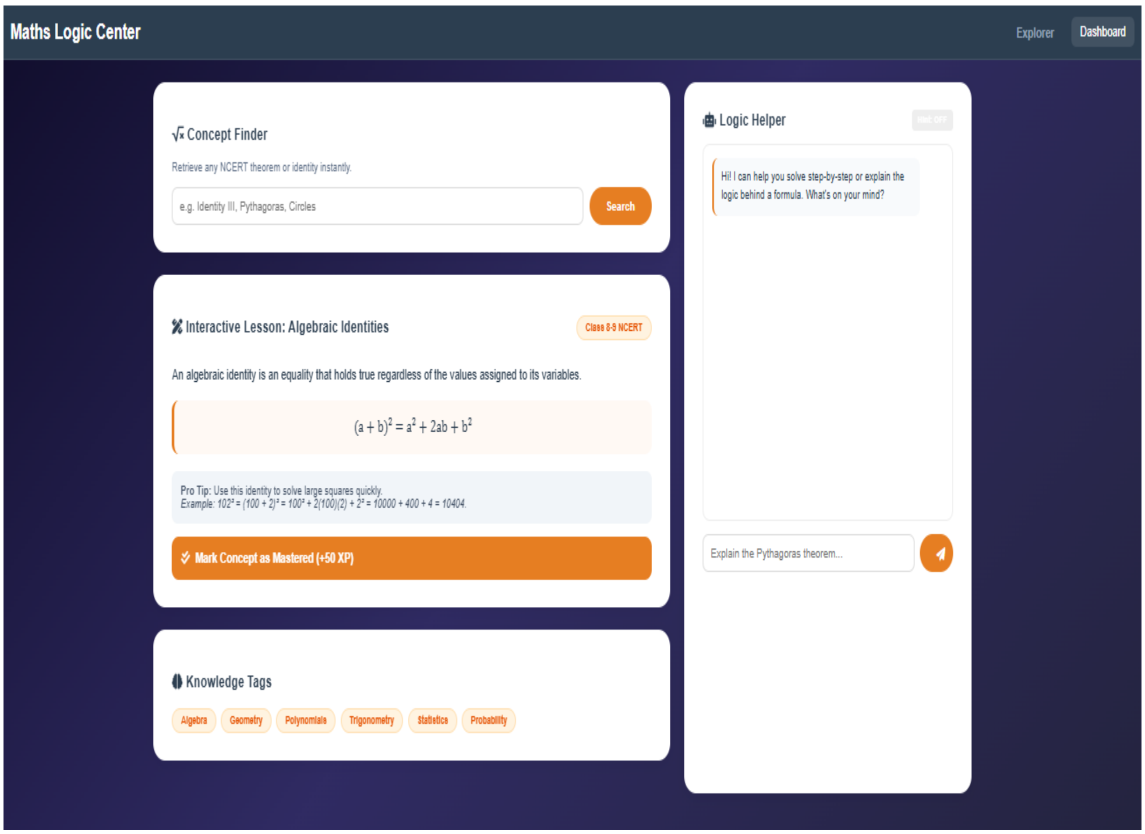
```
    background-color: #ffffff !important;
    color: #2d3436 !important;
}

input:focus {
    border-color: var(--primary);
    outline: none;
    box-shadow: 0 0 12px rgba(108, 92, 231, 0.4);
}

/* 6. Learning Dashboard Layouts */
.content-card, .section-card {
    background: var(--glass);
    backdrop-filter: blur(15px);
    border-radius: 25px;
    border: 1px solid rgba(255, 255, 255, 0.1);
    padding: 30px;
    margin-bottom: 20px;
    text-align: left;
    transition: 0.3s;
}

.content-card:hover {
    background: var(--glass-heavy);
}

/* 7. Buttons */
button, .action-btn {
    padding: 12px 28px;
    background: var(--primary);
    color: white;
    border: none;
    border-radius: 50px;
    cursor: pointer;
    font-weight: bold;
    font-size: 15px;
    transition: 0.3s ease;
    display: inline-flex;
    align-items: center;
    gap: 10px;
}

button:hover {
    background: var(--secondary);
    transform: translateY(-2px);
```

```css
    box-shadow: 0 5px 15px rgba(108, 92, 231, 0.4);
}

/* 8. AI Chat Interface Enhancements */
#chat-display {
    scrollbar-width: thin;
    scrollbar-color: var(--primary) transparent;
}

.msg {
    margin-bottom: 12px;
    padding: 10px 15px;
    border-radius: 15px;
    max-width: 85%;
    line-height: 1.4;
}

.user-msg {
    background: var(--primary);
    align-self: flex-end;
    margin-left: auto;
    border-bottom-right-radius: 2px;
}

.ai-msg {
    background: var(--glass-heavy);
    align-self: flex-start;
    border-bottom-left-radius: 2px;
}

/* 9. Animations */
@keyframes fadeIn {
    from { opacity: 0; transform: translateY(15px); }
    to { opacity: 1; transform: translateY(0); }
}

.login-box, .section-card {
    animation: fadeIn 0.6s ease-out forwards;
}
```

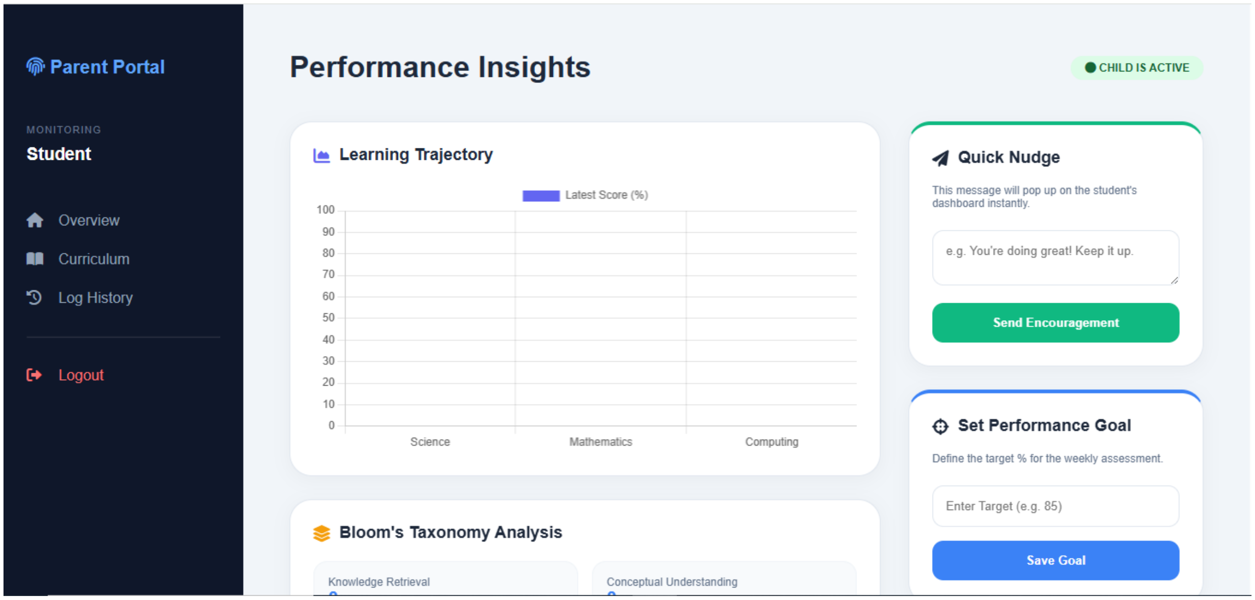
Login Page


APK's Learning
Welcome Back
Student
Parent
Username
Enter your name
Password
Sign In
Password hint: username + 1234

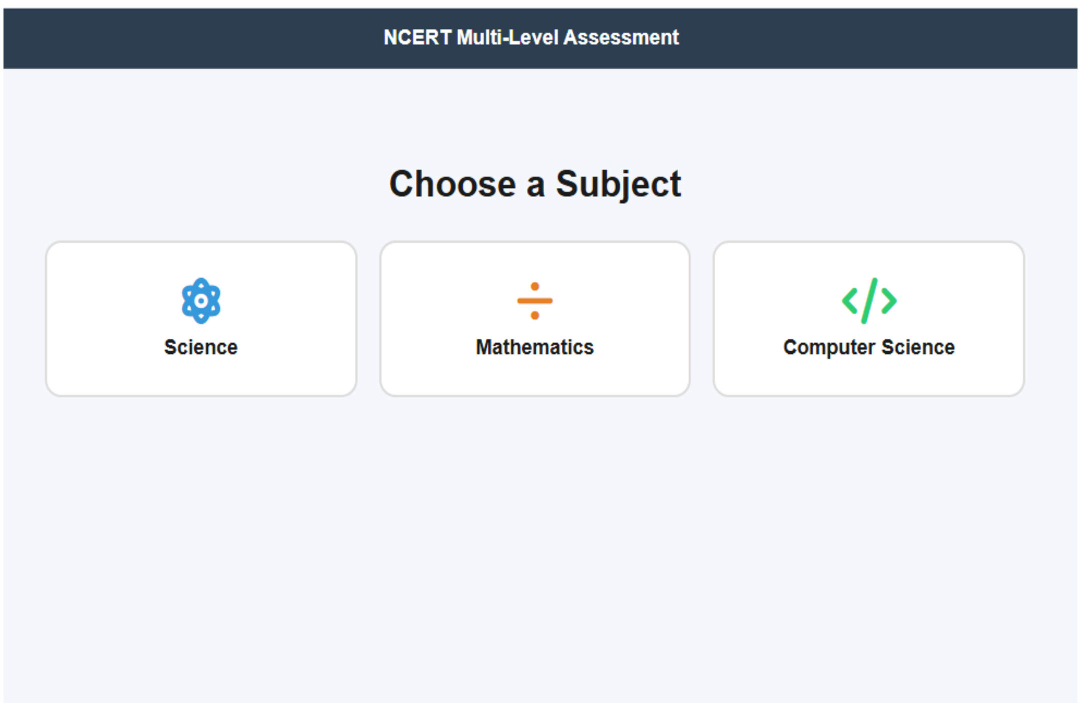
Maths Logic Center



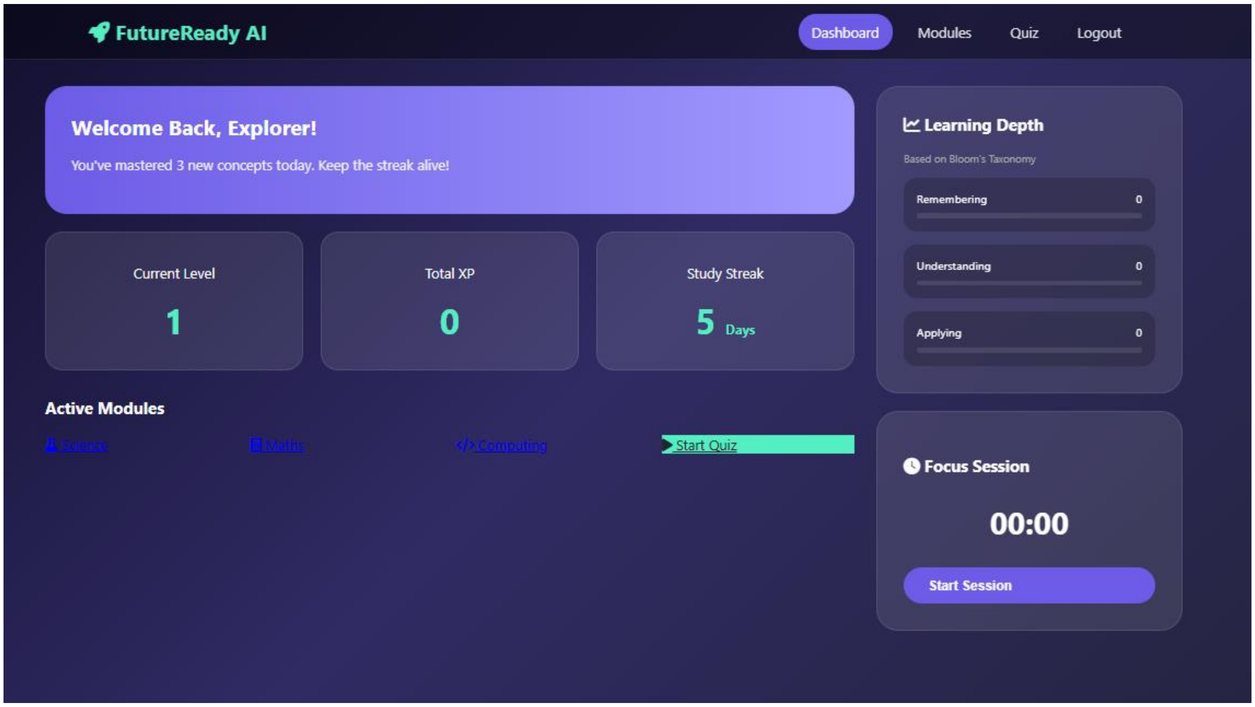
Parent Dashboard



Practice Page



Student Dashboard



Subject Page